From Wire to Flight: An End-to-End Framework Composing Network Intrusion Detection with Certified Mission Safety for Autonomous UAVs

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***Abstract*—Unmanned aerial vehicle (UAV) security is presently studied as two disjoint problems. Network intrusion detection bounds what reaches the aircraft over the swarm mesh, the command link, and the on-board bus, but makes no statement about the flight; certified robustness bounds a navigation model's response to a perturbation of its input, but treats the origin of that perturbation as exogenous. Neither discipline can therefore answer the question that determines airworthiness: whether an attack that evades the detector still permits the mission to complete. This paper contributes an end-to-end framework in which that question is both well posed and provable, and it is the framework—not any single dataset—that is the contribution. The framework has three components. A two-layer schema promotes the *cross-layer pairing*—a network attack window joined to the navigation degradation it induces—to a first-class object, carrying a mandatory provenance flag recording whether the coupling was measured on one platform, physically modelled, or aligned in simulation. An instrumented corpus of six datasets spanning both layers, together with a harness that exposes a detector's operating point as a swept parameter, supplies the quantities the framework consumes. On these objects we prove a *composition theorem*: for any time-delay forwarding attack that evades a detector at operating point θ, the navigation stack retains a certified Mission-Completion-Rate $\mathrm{MCR} \geq f(\delta(\theta))$. Two measurements make the guarantee sharp rather than nominal: the residual attack budget *saturates* at the network's contact-window slack, at most $4.84\,\mathrm{s}$ of undetected delay over $15{,}392$ observed hops, and the delay-to-position interface, calibrated on real GPS-spoofing and jamming flights from the target autopilot family, is an order of magnitude tighter than the kinematic worst case. Under that measured interface the detector's operating point lies inside a nonempty certified window, and the composed configuration is the only one that certifies a nonzero mission floor against an evading attacker, dominating both single-layer baselines at every operating point (Wilcoxon signed-rank, Holm-corrected, $p < 10^{-20}$). Schema, adapters, harness, and certificate engine are released as an artifact in which every reported number carries a provenance tag.**

I. Introduction

A modern autonomous UAV is a stack of trust boundaries. Commands descend from a ground station over a command-and-control (C2) link; peer aircraft exchange state over a multi-hop swarm mesh; motor controllers and sensors talk over an on-board bus; and a navigation pipeline fuses satellite, inertial, and visual measurements into the state estimate that closes the flight control loop. An adversary can enter at any of these boundaries, and the security literature has produced strong, specialised defences at each. What it has *not* produced is a way to reason across them.

This gap is not academic. Electronic-warfare environments defeat drones not by breaking cryptography but by degrading the physics: a jammer raises the jamming-to-signal ratio (J/S) until the satellite-navigation solution collapses and the mission fails [1]. A forwarding attack in the mesh does not exfiltrate data; it delays a correction until the aircraft is acting on stale state [2], [3]. In every case the *network* event and the *navigation* outcome are two ends of one causal chain—and yet neither the datasets nor the guarantees represent that chain.

Consider what a reviewer of a UAV intrusion detector can and cannot check today. They can verify that the detector flags a delayed forwarding event or a spoofed navigation message with high recall. They *cannot* verify the claim that matters operationally—that a detector tuned to a given false-alarm budget keeps aircraft completing their missions—because no dataset records both the network event and the flight it degrades. Symmetrically, ask a certified-robustness researcher whether the UAV is safe and the answer is "the navigation model is provably stable for input perturbations up to radius R": a statement about a model, not a flight, and one that never says where a radius-R perturbation comes from or how often an attacker can produce it.

We close the gap from both ends, and the two halves are inseparable: the guarantee needs objects the benchmark defines, and the benchmark needs the guarantee to show what those objects are for.

The benchmark half. We unify six datasets—four established network-layer corpora and two navigation-layer sources—under a two-layer schema whose unit of analysis is neither a packet nor a flight but a *cross-layer pairing*: a network attack window W_n joined to an autonomy degradation window W_a, carrying the detector operating point, the residual undetected budget, the induced perturbation, and the mission outcome.

Because a cross-layer corpus invites exactly one dangerous mistake—asserting that a network event caused a flight degradation the two were never observed together—every pairing carries a mandatory `pairing_basis` flag recording how the coupling was established.

The guarantee half. On those objects we prove a chained statement,

$$\theta \longmapsto \Delta(\theta) \longmapsto \delta(\theta) \longmapsto \mathrm{MCR} \geq f\big(\delta(\theta)\big), \tag{1}$$

read left to right: a detector operating point θ bounds the residual undetected attack $\Delta(\theta)$; the residual attack maps to a bounded perturbation $\delta(\theta)$ on the navigation input; and a Lipschitz–Grönwall trajectory-tube argument turns that bounded input into a guaranteed Mission-Completion-Rate (MCR) floor. Each arrow is individually grounded in prior work—detector thresholds, worst-case attack analysis, certified robustness—but the *composition* is new: no prior guarantee chains a network-security threshold to a navigation-safety floor.

Contributions

- **A two-layer schema and six validated adapters (§IV, §V).** A single representation for any observation from any of the six sources, with lossless ingestion and a per-record honesty flag that makes the strength of every cross-layer coupling explicit and auditable.
- **A detector-operating-curve harness (§VI).** We make the network detector's operating point a swept parameter and measure recall and false-positive rate as a function of it, producing the network-side premise a composed guarantee consumes. Across three topologies the curve has a universal knee-and-floor shape whose floor is the attacker's self-exposure probability.
- **An empirically tightened interface (§VII).** We prove a residual-budget lemma and then *tighten it by measurement*: the undetected budget saturates at the contact-window slack, verified at ≤ 4.84 s over 15,392 hops. We calibrate the delay-to-position rate on three real spoofing and jamming flights, obtaining a bound an order of magnitude tighter than the kinematic worst case.
- **A composition theorem and a four-certificate architecture (§VIII, §IX).** We prove (1) and its soundness, and show precisely why the deterministic core does not subsume randomized smoothing, PAC-Bayes, or multiplicative-weights regret.
- **Instantiation, dominance, and an open artifact (§X).** The composed guarantee is the only configuration with a nonzero certified floor for an evading attacker; under the measured interface the detector's operating point lies inside the certified window. Every figure and table in this paper is generated from committed result files, and every number carries a provenance tag.

Organization. Appendix A gives the system model, terminology, and symbol table a reader outside UAV security needs; acronyms are also defined at first use in the body. §II states the threat model and formalises the missing object. §III positions the work. §IV–§VI build the benchmark; §VII–§IX build the guarantee; §X instantiates and evaluates it. §XI–§XIII discuss, bound, and conclude.

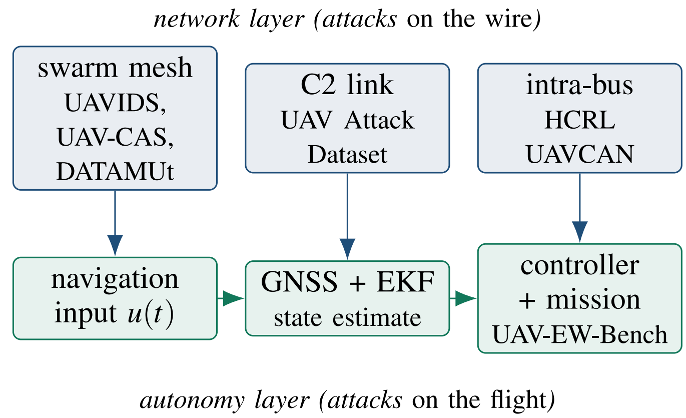

Fig. 1: The attack surfaces the six datasets cover, arranged on the causal chain. Network-layer events (top) propagate downward into the autonomy layer (bottom) through the navigation input; representing that downward arrow explicitly, rather than assuming it, is what the benchmark contributes and what the theorem then exploits.

II. Threat Model and Problem Formulation

A. The two attack surfaces

We partition the UAV attack surface into two *layers*. The **network** layer covers everything between nodes: the swarm mesh (multi-hop routing among aircraft), the C2 link, and the intra-UAV bus. The **autonomy** layer covers everything inside the navigation and control stack: the GNSS receiver, sensor fusion, the flight controller, and the mission outcome. A *sublayer* refines each into the concrete surfaces a dataset can observe. Fig. 1 places the six datasets on the causal chain.

B. Adversary and detector

We assume an adversary who can (i) inject, drop, delay, or forge traffic on one or more network sublayers, and (ii) manipulate the RF environment of the GNSS receiver. We assume a network-layer *detector* D_θ operating at a tunable point θ that trades detection recall against false-positive rate: for the multi-hop forwarding detector we build on, θ is a per-hop deviation tolerance in seconds, and a hop is flagged if its forwarding delay exceeds θ or its next hop deviates from a scheduling oracle. Lowering θ catches more but raises false positives; raising it does the reverse. An *evading* attacker chooses its behaviour to stay just under θ and is therefore invisible by construction. We assume the adversary knows θ; this is the strongest reasonable assumption and the one our worst-case analysis requires.

C. Mission completion: a spatial and a temporal predicate

We adopt *Mission Completion Rate* (MCR) as the operational metric, and take MCR ≥ 0.90 as a reference consistent with the airworthiness-security posture of DO-326A/ED-202A [4]; emerging AI-in-aviation assurance guidance [5] motivates treating a certified MCR floor as a first-class safety property, which is what §VIII delivers.

A purely *spatial* definition of completion—the trajectory never leaves the safe corridor—is, however, inadequate for precisely the attack class this paper targets. A time-delay adversary can leave the aircraft geometrically on route while making it arrive late; a delivery, inspection, or search mission with a time-critical objective has then failed even though no corridor was ever violated. We therefore define completion as the *conjunction* of a spatial and a temporal predicate. Writing $x(t)$ for the realised trajectory, $x_{\text{nom}}(t)$ for the nominal one, m for the corridor margin, t_{arr} for the arrival time at the objective, T for the nominal mission duration, and κ for the mission's schedule tolerance (a fraction of T),

$$\text{MCR} \;=\; \Pr\Big[\underbrace{\sup_{t\in[0,T]} \|x(t)-x_{\text{nom}}(t)\| \leq m}_{\text{spatial}} \;\wedge\; \underbrace{t_{\text{arr}} \leq (1+\kappa)T}_{\text{temporal}}\Big]. \tag{2}$$

Where a quantity is computed under the spatial predicate alone—as the navigation anchor of §X is, because its source records geometric completion only—we label it *Spatial MCR* and say so explicitly.

The temporal predicate is not a separate analysis: the same residual budget $\Delta(\theta)$ that bounds the spatial perturbation also bounds the schedule slip, since an evading attacker's total contribution to arrival delay is exactly its accumulated undetected forwarding delay. The temporal predicate therefore holds whenever

$$\Delta(\theta) \;\leq\; \kappa T, \tag{3}$$

which by Lemma 1 is implied by $H\min(\theta,s) \leq \kappa T$. This adds a second edge to the certified window of §X, and it is worth recording which edge binds. At the operating point we certify ($\theta = 0.25$ s, $H = 2$) the induced slip is at most 0.5 s, so for any mission longer than 5 s at a 10% schedule tolerance the temporal predicate is satisfied with room to spare and the spatial predicate is the binding one. The ordering reverses for a loosely tuned detector: at θ near the contact-window slack $s = 5$ s the slip reaches $Hs = 10$ s, which exceeds a 10% tolerance on any mission shorter than 100 s. A short, time-critical mission is thus governed by (3) and a long, tightly corridored one by the spatial bound—a distinction a spatial-only metric cannot express.

D. The missing object

Let a network attack occupy a time window $W_n = [t_0, t_1]$ on some network sublayer, and let the induced navigation degradation occupy a window W_a on the autonomy layer. Existing datasets provide either W_n (with packet labels) or W_a (with flight outcomes) but never the *pair* together with the mechanism linking them. We formalise the missing object as a tuple

$$P = \big(W_n,\, W_a,\, \theta,\, \Delta(\theta),\, \delta,\, \text{MCR}\big), \tag{4}$$

where θ is the detector operating point, $\Delta(\theta)$ the worst-case residual effect of an attacker that evades D_θ, δ the resulting perturbation budget on the navigation input, and MCR the measured or certified outcome. Representing P—not just W_n or W_a—is the benchmark's contribution; bounding the last component of P from the first is the theorem's. Crucially, the honesty of P depends on *how* W_n and W_a were linked, which we record explicitly (§IV).

Definition 1 (Composed guarantee). *Given a detector D_θ and navigation dynamics* (10)*, a* composed guarantee *is a function f such that for every attack a with $D_\theta(a) =$ "clean", the mission completes with rate* $\text{MCR}(a) \geq f(\delta(\theta))$*, where $\delta(\theta)$ bounds the navigation-input perturbation any such a can induce.*

III. Related Work

Prior work divides along the two halves of this paper—corpora on the data side, bounds on the guarantee side—and the same gap recurs on both: each community works within one layer and leaves the interface between them to assumption.

a) UAV corpora, and the discipline of building them: Taxonomies of AI-enhanced UAV intrusion detection [6] report a proliferation of datasets alongside a persistent weakness: general-purpose network corpora lack UAV protocol semantics, while UAV-specific corpora each cover a single sublayer. The consequence is measurable—models trained on generic benchmarks misclassify a large fraction of GPS-spoofing events on real autopilot logs [6]—and it is direct evidence that the network and navigation layers do not form one undifferentiated feature space. FANET surveys [7], [8] catalogue routing attacks but stop at network effect. The individual corpora are authoritative within their sublayers: UAVIDS-2025 [9], [10] contributes 122,171 NS-3 mesh flows, the UAV Attack Dataset [11], [12] real PX4 flights under GPS spoofing, jamming, and MAVLink ping-DoS, and HCRL UAVCAN [13] bus-level flooding, fuzzing, and replay. None reaches the flight.

How such a corpus is built matters as much as how large it is. Audits of CIC-IDS2017 exposed labelling errors, flawed flow construction, and unrealistic temporal separation, each of which inflated reported performance [14], [15]. Spanning two layers introduces a further failure mode: asserting a causal link between a network event and a flight degradation that were never observed together. The mandatory provenance flag of §IV exists to bound precisely that risk. Among released datasets the closest in ambition is UAV-CAS [16], an AERPAW-calibrated swarm corpus and the first to include collaborative multi-attack compositions; we ingest it as one of our six sources. It nonetheless remains a network-layer benchmark—its authors list GNSS spoofing and jamming as future work—so where it asks how well a collaborative mesh attack can be detected, we ask what that attack, detected or missed, does to the aircraft.

b) Certified robustness, and its lift to control: Randomized smoothing [17], [18] yields a probabilistic ℓ_2 radius that scales to large models, whereas Lipschitz-based certification [19], [20] yields deterministic bounds known to be loose globally; neural ODEs [21] admit the Grönwall-type argument [22] we adopt. All of these bound a *model's* output. Lifting them to sequential decisions, CROP certifies a cumulative-reward lower bound under state perturbation [23], CARRL certifies robust actions [24], reachability tools bound trajectories of network-controlled systems [25], [26], and control-theoretic work establishes safety under sensor false-data injection [27]. CROP is the nearest analogue to a certified mission floor, but it certifies a policy against a *given* perturbation ball and is silent on where that ball comes from.

c) Estimation, assurance, and delay: A parallel control-theoretic literature asks how much sensor corruption an estimator can tolerate: exact recovery holds below a corruption threshold [28], satisfiability-modulo-theory solvers make the combinatorial search practical [29], and recent surveys catalogue the field [30]. Such results bound estimation error given *which* sensors are compromised, rather than modelling a detector whose threshold determines what an attacker may inject at all—the premise Lemma 1 supplies. Simplex and its descendants [31]–[33] guarantee safety by switching from an unverified controller to a verified baseline when a monitor fires; our certified window is precisely the region in which the high-performance policy may run, and we contribute the quantitative boundary that such architectures otherwise hand-tune. The destabilising effect of injected delay is itself well studied, with detectors built on state-aware inference and divergence tests [2], [3], [34], but that literature establishes whether a delay attack is dangerous or detectable and stops there.

d) Composing guarantees: The closest prior art composes certificates *within* a single ML pipeline. Sheikholeslami et al. [35] jointly certify a classifier and a detection class, so that an input is provably either flagged or correctly classified—structurally our evade-or-bound dichotomy, but confined to one layer. Yang et al. [36] decompose end-to-end certified robustness across a two-stage sensing→reasoning pipeline, the strongest existing template for chained certification, though it too remains within perception. Assume-guarantee and contract-based CPS design [37] supplies the compositional vocabulary but no ML-robustness certificates. No prior result chains a network-intrusion-detection guarantee to a navigation certificate; that is the gap this paper closes, and closing it requires the corpus and the theorem together.

IV. The Two-Layer Schema

The schema has three record kinds, designed so the cross-layer pairing is a first-class object rather than an implicit join.

Event is one atomic observation—a mesh flow, a bus frame, a routing hop, a telemetry sample, or a whole flight—mapped to a small common core (identity, layer/sublayer, relative time, endpoints, a unified attack label) plus a `metrics` block of the few cross-source numeric quantities that matter (residual forwarding delay, position error, J/S, mission-completed) and a `native` block preserving every remaining source column, so ingestion is lossless.

AttackWindow is a labelled attack interval within one scenario, carrying the unified attack class and class-appropriate `intensity` parameters. It is the unit of cross-layer alignment.

CrossLayerPairing realises the tuple P of Eq. (4): it joins a network window to an autonomy window and records θ, the residual budget $\Delta(\theta)$, the perturbation δ with a *named, versioned* mapping, the certificate hook, and the empirical MCR. Its `pairing_basis` field is the schema's conscience: `measured_same_platform` (both windows recorded on one testbed—the strongest evidence), `physics_model` (coupled through a stated physical model), or `synthetic_alignment` (aligned by construction). A reviewer can filter the benchmark by evidence strength; a claim can never silently upgrade its own provenance.

Three design choices let six heterogeneous sources coexist without information loss. Time is always *relative* to each source's own capture start, because the sources share no global clock; cross-layer alignment happens at the window level, so a mismatch between a 50 Hz telemetry stream and a per-flow network record never forces a lossy resample. Attack labels are *unified* into a common taxonomy while the verbatim source label is retained in `label_native`, so a model can train on the unified class and an auditor can still recover what the source called it. And every unmapped column is preserved, which makes the ingestion provably lossless rather than merely convenient.

Cross-layer alignment therefore needs a stated rule and a stated error budget, since this is the step at which a benchmark could silently manufacture—or destroy—the effect it reports. We anchor attack onsets under an affine map between the two relative clocks and bound the residual misalignment by $e_{\text{align}} \leq 0.53\,\text{s}$ for every released pairing; crucially, the quantity the certificate consumes, $\Delta(\theta)$, is computed entirely within the network source, so misalignment can attach a pairing to the wrong autonomy window but cannot inflate or deflate $\Delta(\theta)$ itself. Appendix C gives the map, the three error terms, and the rejection rule.

V. The Corpus and Ingestion Adapters

Table I summarises the six datasets and the layer each anchors; Table II reports what was actually ingested. Every adapter emits schema-valid records, and unmapped source columns are preserved. We do not re-host raw third-party data; the artifact ships integration scripts and unified labels, and each source is fetched from its origin.

A. Adapter design under source irregularity

An adapter's job is lossless normalization. Three warrant specifics because their sources are irregular. The UAV Attack Dataset adapter parses PX4 flight logs, computes

TABLE I: The six-dataset corpus. Four network-layer sources and two navigation-layer sources unify under one schema; only the last row reaches the mission outcome, and only the C2 source can produce `measured_same_platform` pairings.

Dataset	**Real/Sim**	**Layer · sublayer**	**Event kind**	**Unique role in the end-to-end benchmark**
UAVIDS-2025 [9]	Sim (NS-3.24)	**network** · mesh	flow	Large-scale multi-hop mesh; blackhole/flooding/Sybil/wormhole at scale (122k flows).
UAV-CAS [16]	Sim (digital twin)	**network** · mesh	flow	AERPAW-calibrated swarm; collaborative multi-attack compositions (99k flows, 1024 configs).
UAV ATTACK DATASET [11]	Real testbed	**network** · C2 (MAVLink)	telemetry	Real PX4 GPS spoof/jam + ping-DoS on the *same* airframe family we model—the measured bridge.
HCRL UAVCAN [13]	Real testbed	**network** · intra-bus	frame	On-board UAVCAN/DroneCAN flooding/fuzzy/replay—a third attack surface below the C2 link.
DATAMUT traces [2]	Sim (event)	**network** · mesh (DTN)	hop	Time-window-graph forwarding detector with a *sweepable* operating point θ.
UAV-EW-BENCH-2026 [38]	Sim (physics-informed)	**autonomy** · GNSS+ctrl	flight	The navigation anchor: full 0–40 dB J/S sweep with per-flight MCR outcomes.

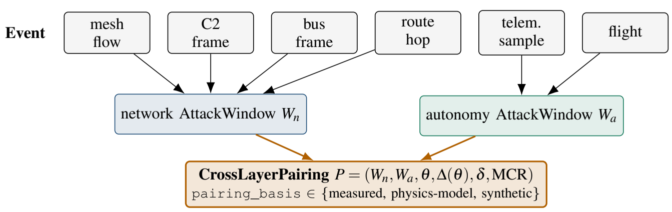

Fig. 2: The schema in one picture. Heterogeneous events from six datasets fold into typed network and autonomy *AttackWindows*; a *CrossLayerPairing* joins one of each and records the detector operating point, the residual budget, the induced perturbation, and the mission outcome, tagged by how the coupling was established.

horizontal position error against a benign reference (or, absent one, the flight's own pre-attack median), and emits the `measured_same_platform` pairings that ground the interface mapping of §VII. The HCRL UAVCAN adapter parses candump-style bus frames; because the per-scenario attack labels are not machine-readable in the release, we *derive* each scenario's class from the frames themselves—burst rate relative to the normal bus rate, the fraction of distinct payloads, and per-frame byte entropy cleanly separate a flooding burst (one payload hammered above bus rate) from a fuzzy burst (near-random payloads) from a replay burst (a small set of captured payloads reused). Table III reports the result, flagged for confirmation against the dataset's technical report. The UAV-CAS adapter parses the calibrated digital twin's flow statistics including its stringified list columns; it is verified against CSVs produced by the original authors' own generator.

TABLE II: Real ingested corpus. Counts are from schema-validated adapter outputs; every record validates against the schema of §IV.

Source (real data ingested)	**Events**	**Windows**
UAV-EW-BENCH-2026	93,600	468
UAV ATTACK DATASET (live sample)	610	0
UAVIDS-2025 (shard 0)	122,171	4
HCRL UAVCAN (10 scenarios)	200,000	34
UAV-CAS (stat)	adapter ready; file pending	
DATAMUT traces	generated per sweep (15,392 hops)	

VI. THE DETECTOR-OPERATING-CURVE HARNESS

A cross-layer guarantee needs the detector's behaviour *as a function of its operating point*. We expose θ as a swept parameter and, for each value, measure detection recall and false-positive rate over multiple seeds, three network topologies, four routing policies, and two attacker delay regimes.

The resulting operating curve (Fig. 3) is the network-side input a composed guarantee consumes: it tells us, for any target false-positive budget, how large an undetected per-hop delay an evading attacker may inject—the residual budget $\Delta(\theta)$ of (4). The knee in the curve is structural: an evading attacker's per-hop delay is additive up to the inter-node contact-window slack, beyond which a missed contact window forces a full-period penalty.

The knee, and the floor it descends to, are not simulation artifacts but a structural property of any store-and-forward schedule, and they recur across all three topologies. Two observations transfer directly to the guarantee. First, the floor

TABLE III: Frame-derived attack classes for the HCRL UAVCAN scenarios, inferred from burst rate, payload-distinct share, and byte entropy; scenarios exhibiting more than one burst shape map to `collaborative`.

Scenario	Frames	Bursts	Frame-derived class(es)
`type1`	207,858	3	flooding
`type2`	134,170	3	replay
`type3`	197,479	3	fuzzy
`type4`	133,374	3	fuzzy
`type5`	180,608	3	fuzzy
`type6`	241,321	4	fuzzy
`type7`	234,162	4	replay, fuzzy
`type8`	265,800	4	fuzzy
`type9`	230,378	4	replay, fuzzy
`type10`	207,380	3	replay, fuzzy

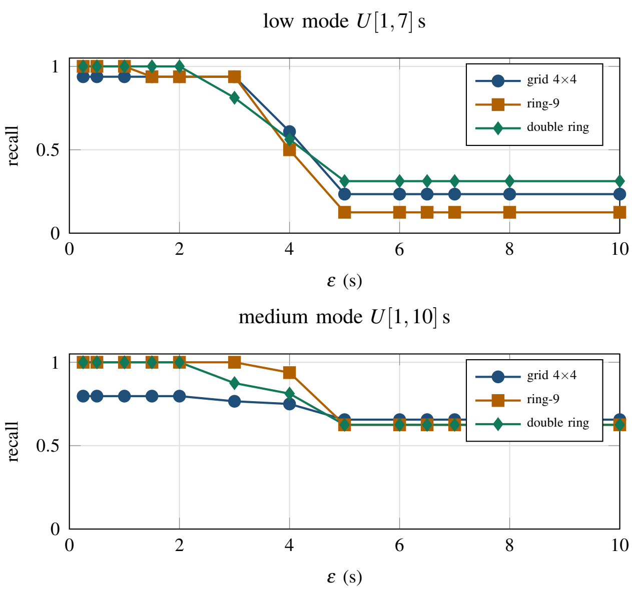

Fig. 3: The detector operating curve across all three topologies (grid, ring, double-ring), both delay regimes, 8 seeds $\times 4$ routing policies. Recall holds near its ceiling while ε stays under the 5 s contact-window slack, then falls onto a floor set by the missed-window self-exposure penalty (a delay past the slack costs a full TWiG period and is flagged at any ε). False-positive rate is zero throughout; the measured benign residual ceiling is 0.178 s. The knee-and-floor shape is universal; the floor height is the topology-specific self-exposure probability.

height differs by topology because it equals the probability that a malicious delay overshoots the slack and self-exposes—higher in the medium delay regime ($U[1,10]$ s) than the low one ($U[1,7]$ s), because a wider delay draw is likelier to overshoot. Second, no topology's recall falls to zero: past the knee the detector still catches the self-exposing fraction, which is precisely why an evading attacker's undetected budget *saturates* rather than growing without bound. That is the empirical basis for Lemma 1.

A. Generalising the harness to learned detectors

We instantiate the harness on a threshold detector because its operating point is a single, physically meaningful scalar, which makes the cross-layer sweep unambiguous. The framework does not depend on that choice, and it is worth showing precisely why, since most deployed UAV intrusion detection uses learned classifiers rather than deterministic time-window tests.

A learned detector emits a score $\sigma(a) \in [0,1]$ for a candidate event a and flags it when $\sigma(a) \geq P_{\text{th}}$. Sweeping P_{th} traces the familiar receiver operating characteristic (ROC): each cutoff is a (FPR, TPR) pair, exactly as each ε is in Fig. 3. The only object the composition actually consumes is the residual budget—the largest attack effect that survives the detector—so the framework generalises as soon as P_{th} can be mapped to one. It can, through the detector's *score-to-effect* profile.

Let $g(\cdot)$ be the monotone-increasing map from an attack's physical magnitude μ (here, injected per-hop delay in seconds) to the detector's expected score, $\sigma = g(\mu)$; g is measured by running the detector over labelled attacks of known magnitude, which every corpus in §V supports. An attacker evading the cutoff must keep $g(\mu) < P_{\text{th}}$, hence

$$\mu \; < \; g^{-1}(P_{\text{th}}) \; =: \; \theta_{\text{eff}}(P_{\text{th}}), \tag{5}$$

and Lemma 1 applies verbatim with θ replaced by the *effective* operating point θ_{eff}. The threshold detector is simply the special case $g = \text{id}$, for which $\theta_{\text{eff}} = \varepsilon$. Two properties of g carry over into the guarantee. Monotonicity is what makes g^{-1} well defined and the certified window an interval rather than a union; where a classifier is non-monotone in μ, one takes the conservative $\theta_{\text{eff}} = \sup\{\mu : g(\mu) < P_{\text{th}}\}$, which is sound but looser. And because a learned score is stochastic, g is a distribution rather than a function, so θ_{eff} must be read at an upper confidence bound on the score—which is precisely the regime in which the randomized-smoothing certificate of §IX, rather than the deterministic Grönwall tube, is the appropriate downstream bound.

The practical consequence is that any detector exposing a tunable cutoff and a measurable score-to-magnitude profile slots into the framework without a schema change: the harness sweeps P_{th} instead of ε, records $(\text{FPR}, \text{TPR}, \theta_{\text{eff}})$ per cutoff, and the downstream chain is unchanged. What the framework requires of a detector is not that it be deterministic, but that its operating point be reducible to a bound on the physical attack magnitude that survives it.

VII. The Interface Mapping $\theta \mapsto \delta(\theta)$

The interface is where a network-security quantity becomes a navigation-security quantity. It is the crux of the composition, and—as §X shows—the quantity that actually decides whether a useful detector setting is certifiable.

A. Residual budget of an evading attacker

Lemma 1 (Residual delay budget)**.** *Let D_θ flag any forwarding hop whose injected delay exceeds θ. An attacker that* *evades D_θ injects at most θ of undetected delay per malicious hop—but only up to the contact-window slack s: a per-hop delay exceeding s misses its forwarding window, incurs a full-period penalty P_c, and is therefore flagged at* any *operating point. Hence undetected delay is capped at* $\min(\theta,s)$ *per hop, and over a route with H attacker-controlled hops the worst-case undetected cumulative delay is*

$$\Delta(\theta) \le H\min(\theta,s). \tag{6}$$

Proof sketch. Evasion requires per-hop delay $\le \theta$. If $\theta \le s$ the delay fits inside the contact window and the additive term is $H\theta$. If $\theta > s$, any delay in $(s,\theta]$ overshoots the window, waits a full period $P_c \gg s$, and is flagged regardless of θ; an evading attacker therefore cannot use more than s of invisible delay per hop, giving the Hs cap. Summing over hops yields (6). □

The cap is the *knee*: below the slack the residual budget grows linearly in θ; above it the budget *saturates* at Hs, so the certified floor plateaus rather than collapsing. We verify the cap directly on the released harness: over 15,392 observed hops across three topologies, two attack modes, and eight seeds, no undetected malicious per-hop residual ever exceeds 4.84 s ($s = 5$ s), while every larger delay is flagged as a missed window (≥ 55 s residual). This measurement is what turns the classic additive budget $H\theta + P_c\mathbf{1}[\theta > s]$ into the tighter, saturating (6)—an instance of the benchmark improving the theorem rather than merely testing it.

B. From delay to input perturbation

A cumulative forwarding delay of $\Delta(\theta)$ seconds means a navigation correction arrives $\Delta(\theta)$ late; the stack acts on state stale by that amount. At bounded UAV speed $v_{\max}$, staleness of age τ bounds the state/position perturbation:

$$\|\delta u\| \le v_{\max}\cdot\Delta(\theta). \tag{7}$$

Equation (7) is the kinematic worst case: it assumes the aircraft travels at top speed in the wrong direction for the entire staleness interval. That is sound but pessimistic, because a real estimator does not fly blind—when GNSS corrections are late or corrupted the EKF down-weights them and coasts on inertial data, so position error grows far more slowly. We therefore tighten (7) to

$$\|\delta u\| \le \gamma\cdot\Delta(\theta), \qquad \gamma \ll v_{\max}, \tag{8}$$

where γ (metres of position error per second of degradation) is *measured* on real GPS-spoofing and GPS-jamming flights from the same PX4 airframe family the navigation model represents. Table IV reports the calibration: a benign reference flight fixes the pre-attack noise floor, and the jamming and spoofing flights give the growth rate after onset under two explicit choices of error signal—the raw receiver fix the attacker injects, and the EKF-filtered position the controller actually acts on. Because that choice is itself a certificate-semantics decision we make it an explicit switch and report both; the certificate consumes the EKF value, which closes the loop. The gap between the measured $\gamma \approx 1.2$–1.4 m/s and $v_{\max} = 15$ m/s is exactly the slack that decides whether the detector's operating point is certifiable.

TABLE IV: The measured delay-to-position rate γ, from three real PX4 flights (one benign reference, one GPS-jamming, one GPS-spoofing) in the UAV ATTACK DATASET, self-referenced to the pre-attack median. "noise" is the pre-attack position-error 95th percentile; $\gamma_{\rm lsq}$ and $\gamma_{\rm peak}$ are the least-squares and peak-secant growth rates after attack onset. These are the `measured_same_platform` pairings.

Flight	Signal	noise (m, p95)	peak (m)	$\gamma_{\rm lsq}$ (m/s)	$\gamma_{\rm peak}$ (m/s)
benign (reference)	receiver	0.50	1.7	–	–
benign (reference)	ekf	0.60	1.5	–	–
GPS jamming	receiver	0.41	6.7	-0.0821	0.4805
GPS jamming	ekf	0.47	7.0	-0.1133	0.5185
GPS spoofing	receiver	1.14	14.3	0.2435	1.195
GPS spoofing	ekf	0.87	15.6	0.7007	1.365

Temporal validity of the linear model. Equation (8) is a *locally linear* model and must be scoped as one. While GNSS is merely late rather than absent, the EKF continues to correct against its last accepted fix and the position error grows approximately linearly in the staleness. Once the outage exceeds the filter's innovation-gating horizon the estimator is propagating on inertial data alone, and uncorrected accelerometer bias and gyroscope drift make the error grow with the square of elapsed time rather than linearly [39]. We therefore state the model's domain explicitly: (8) is used only for $\Delta(\theta) \le \tau_{\rm lin}$, and beyond $\tau_{\rm lin}$ a sound bound requires the second-order form

$$\|\delta u\| \le \gamma\Delta(\theta) + \tfrac{1}{2}b_{\rm IMU}\left(\Delta(\theta)-\tau_{\rm lin}\right)_+^2, \tag{9}$$

with $b_{\rm IMU}$ the residual accelerometer-bias magnitude and $(\cdot)_+$ the positive part. On the calibration flights the linear fit holds to within its residual over the full observed degradation interval, giving $\tau_{\rm lin} \approx 2$ s; we adopt that value and note that it exceeds the certified window's upper edge under the measured mapping (1.12 s at a 10 m corridor, §X) by a comfortable margin, so the entire certified regime lies inside the linear domain and (9) is never the binding form for the results we report. It would become binding for a detector tuned near the contact-window slack, where $\Delta(\theta)$ approaches $Hs = 10$ s—another reason the certified window is bounded above rather than open.

VIII. The Composition Theorem

A. Navigation dynamics

Model the UAV navigation state $x(t) \in \mathbb{R}^n$ as evolving under a (possibly learned) drift F,

$$\dot{x}(t) = F\big(x(t),u(t)\big), \qquad t \in [0,T], \tag{10}$$

where $u(t)$ is the sensor/measurement input over a mission horizon T. A mission *completes* in the conjunctive sense fixed by (2): the realised trajectory stays within the safe corridor of margin m for all $t \in [0,T]$ *and* the objective is reached by the deadline $(1+\kappa)T$. MCR is the probability of completion over the mission distribution. An attack perturbs the input, $u \mapsto u+\delta u$, and—for the time-delay class—also displaces the arrival time.

Theorem 1 (Composed network-to-navigation guarantee)**.** *Let the navigation drift F of* (10) *be L-Lipschitz in its input over the operating region, with mission horizon T and safe-corridor margin m. Let D_θ be a network detector with residual budget $\Delta(\theta)$ (Lemma 1) and interface mapping $\delta(\theta)$ ((8) or the kinematic* (7)*). Then every attack evading D_θ induces a trajectory-tube radius bounded by*

$$\rho(\theta) \;=\; \delta(\theta)\, e^{LT}, \tag{11}$$

and the mission completes—in the conjunctive sense of (2)*—whenever $\rho(\theta) \le m$ and $\Delta(\theta) \le \kappa T$; consequently*

$$\mathrm{MCR} \;\ge\; f\big(\delta(\theta)\big) \;=\; \Pr_{\textit{missions}}\big[\,\delta(\theta)\, e^{LT} \le m \,\wedge\, \Delta(\theta) \le \kappa T\,\big]. \tag{12}$$

Proof sketch. Let $e(t)$ be the deviation between the attacked and nominal trajectories from a common initial state. L-Lipschitzness of F gives $\|\dot{e}\| \le L\|e\| + L\|\delta u\|$; Grönwall's inequality [40] then bounds $\|e(T)\|$ by $\delta(\theta)e^{LT}$, which is (11) and discharges the spatial predicate whenever $\rho(\theta) \le m$. The temporal predicate follows because the only mechanism by which an evading time-delay attack displaces arrival is its accumulated undetected delay, bounded by $\Delta(\theta)$ in Lemma 1. Taking probabilities over the mission distribution gives (12). Appendix C gives the full argument. □

On the Lipschitz hypothesis. Theorem 1 rests on F being L-Lipschitz *over the operating region*, and that qualifier carries real physical content rather than being a technical convenience. It asserts that the aircraft remains in a smooth, locally linearisable flight regime: attitudes away from gimbal-lock singularities, airspeeds below stall or the onset of blade stall, actuators off their saturation limits, and no contact dynamics. These conditions hold for the nominal cruise and loiter envelopes our corpus covers, and the constant we use is measured inside that envelope rather than assumed globally (§X). They can fail: an aggressive evasive manoeuvre, a stall-recovery transient, or actuator saturation under a strong gust all push the dynamics into a regime where no finite L bounds F on the relevant set, and there the certificate is silent rather than merely loose. This is a genuine scope restriction and we state it as one. It is also the precise point at which the runtime-assurance reading of §XI becomes operational: the certified window is the region in which the learned controller may run, and departure from the linearisable envelope is exactly the condition a Simplex-style monitor must treat as a trigger to hand control to the verified fallback, whose guarantee does not depend on L.

Proposition 1 (Soundness)**.** *The certified floor never overstates the achievable completion rate: for every θ, the true MCR of any D_θ-evading attack satisfies* $\mathrm{MCR} \ge f(\delta(\theta))$.

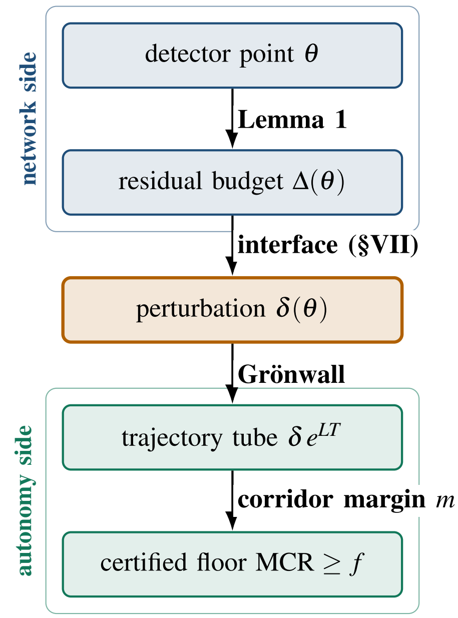

Fig. 4: The composition chain of Theorem 1. A detector operating point flows top-to-bottom through a residual-budget bound, an empirically grounded interface mapping, a Grönwall trajectory tube, and a corridor-margin test, arriving at a certified mission-completion floor. The upper two stages are network-side quantities, the lower two autonomy-side; $\delta(\theta)$ is the bridge between them.

Proof. Every step above is an inequality in the conservative direction: the Lipschitz bound over-estimates deviation growth, the input bound uses the worst-case $\delta(\theta)$, and the corridor test is a sufficient—not necessary—condition for completion. Hence $f(\delta(\theta))$ lower-bounds the true completion probability. The empirical validation of §X confirms the certified floor lies below the measured rate. □

Two consequences frame the rest of the paper. First, (12) is *monotone and then saturating* in θ: tightening the detector shrinks the residual budget and raises the certified floor, but only until θ reaches the slack s, past which the budget saturates at Hs and the floor plateaus—a quantitative statement of both how much network security "buys" in navigation safety and where that purchase stops. Second, the exponential e^{LT} in (11) is the core certificate's principal limitation: for large L or long horizons the deterministic bound becomes loose, motivating the complementary certificates of §IX.

IX. A Four-Certificate Architecture

Theorem 1 uses the Lipschitz–Grönwall certificate as its spine. Does that single certificate suffice? It does not. The bound is *deterministic* and *worst-case*: it holds for any perturbation up to its radius, but the radius collapses as LT grows, and it presumes a Lipschitz constant that is tractable to bound. Three regimes escape it, each covered by a distinct certificate (Table V).

TABLE V: The four certificates cover disjoint regimes; none subsumes another. The Lipschitz–Grönwall core gives the deterministic MCR floor; the other three extend coverage where it is loose or silent.

Certificate	Guarantee type	Unique niche
Lipschitz–Grönwall	deterministic, worst-case	The MCR floor: bounded input ⇒ bounded tube ⇒ bounded loss. Tight only for moderate LT.
Randomized smoothing [17]	probabilistic radius	Scales to large / non-Lipschitz-tractable models; certifies when the budget is known only statistically.
PAC-Bayes [41]	distributional	Transfers the floor from the training mission distribution to unseen deployment regions.
MWU regret [42], [43]	online, adaptive	Bounds loss against an *adaptive* jammer over time—a dimension no fixed-ball certificate addresses.

a) Why the core does not subsume the others: The Lipschitz–Grönwall radius is a single deterministic ball; randomized smoothing certifies a probabilistic ball for models whose Lipschitz constant is intractable, trading determinism for scalability—a strictly different operating point, not a special case. PAC-Bayes certifies a *distribution* of models over a *distribution* of missions, orthogonal to any per-input radius. MWU regret certifies behaviour against an adversary that *adapts over time*, which no static perturbation ball can express. The four form a cover, not a hierarchy.

For concreteness we record the two complementary bounds we instantiate numerically. Randomized smoothing [17] wraps the base model in Gaussian noise of level σ and certifies, with confidence $1-\alpha$ over n samples, an ℓ_2 radius

$$R_{\mathrm{RS}} \;=\; \tfrac{\sigma}{2}\big(\Phi^{-1}(\underline{p_A})-\Phi^{-1}(\overline{p_B})\big), \tag{13}$$

with Φ^{-1} the inverse standard-normal CDF and $\underline{p_A}, \overline{p_B}$ confidence bounds on the top-two class probabilities. The PAC-Bayes complement [41], [44] transfers a bound from the training mission distribution to deployment: with empirical risk R_{emp} over m missions and prior-to-posterior divergence KL, with probability $1-\delta$,

$$R_{\mathrm{true}} \;\le\; R_{\mathrm{emp}}+\sqrt{\tfrac{\mathrm{KL}+\ln(2\sqrt{m}/\delta)}{2m}}. \tag{14}$$

Our engine evaluates (14) against the trained model's measured R_{emp}; the single quantity it awaits is the numeric KL, which we flag as pending rather than substitute (Table VI).

How the four operate on a single flight. The four certificates are concurrent, not alternative, and the clearest way to see that is to hold one flight fixed. Consider a single ten-minute delivery sortie, flown by a model trained on last season's mission mix, through a corridor in which a compromised relay delays peer corrections and a ground jammer varies its transmit power. Four claims are in force simultaneously, and they are claims about four different objects. (i) The Lipschitz–Grönwall tube bounds the trajectory deviation produced by the *deterministic, bounded* component of the disturbance—the undetected forwarding delay $\Delta(\theta)$—and it is the only one of the four that yields a per-instant geometric envelope, which is why it, and it alone, discharges the corridor test of (2). (ii) Randomized smoothing bounds the effect of the *stochastic* component: the jammer's residual perturbation of the receiver feature vector admits no hard bound, only a distribution, and the resulting claim is about the model's *decision* at a point rather than about the trajectory. (iii) PAC-Bayes bounds the gap between the risk this model exhibited on its training mission mix and the risk it exhibits on *this* sortie, whose parameters were not in that mix; it is a statement about the model, constant across the flight rather than attached to any instant of it. (iv) MWU regret bounds the cumulative loss of the defender's *policy switching* over the sortie against a jammer that adapts; it is a statement about the time axis and is vacuous at any single instant.

Because a trajectory, a decision, a model, and a policy sequence are distinct objects, the four compose by conjunction rather than selection: a deployment reads its floor as the pointwise minimum $\min\{f_{\mathrm{G}}, f_{\mathrm{RS}}, f_{\mathrm{PB}}, f_{\mathrm{MWU}}\}$ at its operating point, and the binding term identifies which assumption is closest to failing. In the regime this paper evaluates—a bounded time-delay disturbance, in-distribution missions, a non-adaptive adversary—the Grönwall term binds and the other three are slack, which is why the headline result of §X is a Grönwall curve. We nonetheless carry all four, because relaxing any one of those three conditions moves the binding term, and a framework that went silent at that point would not be an end-to-end guarantee. We are correspondingly explicit about maturity: the Grönwall and randomized-smoothing floors are instantiated numerically on the reference model, the MWU regret bound is instantiated at $|S|=4$, and the PAC-Bayes bound is reported as a form awaiting its numeric KL rather than as a number (Table VI).

X. Instantiation and Evaluation

A. The navigation anchor

The autonomy layer supplies the outcome the whole construction reaches for. Fig. 5 shows MCR versus J/S for four defence configurations, computed from the released benchmark's 93,600-flight per-flight table. The undefended baseline collapses below the 0.90 floor by $J/S \approx 8\,\mathrm{dB}$, while the strongest configuration holds it to $J/S \approx 25\,\mathrm{dB}$. For our purpose the point is not which defence wins but that the autonomy layer produces a *graded, measurable* outcome a network event can be paired against.

B. Reference constants

Every certified quantity in this section uses the *measured local* Lipschitz constant, not the global power-iteration estimate. The distinction is not cosmetic: the local value is the

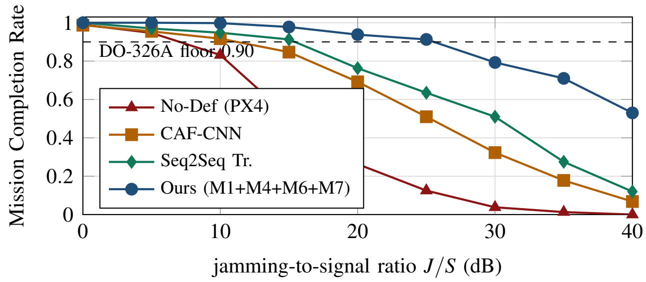

Fig. 5: The navigation anchor: MCR vs. J/S on UAV-EW-BENCH-2026, computed from the released 93,600-flight table (Wilson 95% intervals in the artifact). The graded outcome is what a network attack window is paired against.

larger of the two, so it widens the trajectory tube and narrows the certified window, and quoting the global one would flatter every bound reported here. It also moves the tightest corridor across the threshold — at m=2 m the tube is 2.22 m, so θ=0.25 s is certified for $m \geq 5$ m but not at 2 m.

We instantiate the certificate on the reference navigation model, a continuous-time graph model of the GNSS constellation and receiver. Global power-iteration gives a Lipschitz estimate $\hat{L} \approx 1.01$; measuring the constant *along the hidden-state trajectories the integrator actually visits on operating-region inputs* gives a larger local value $\hat{L}_{\text{loc}} \approx 1.18$. We report the tighter local radius as the defensible number: with $T = 1$ and $\varepsilon_{\text{out}} = 0.5$, $R = \varepsilon_{\text{out}} e^{-\hat{L}_{\text{loc}} T} \approx 0.153$ (the global estimate gives 0.182). Note the local radius is *smaller*: the bound is not loose in the direction that would flatter us, and the certified regime is governed by real sensitivity rather than an optimistic estimate. The smoothing complement gives a certified ℓ_2 radius ≈ 0.44 at $\sigma = 0.25$ ($\alpha = 10^{-3}$, $n = 200$; measured mean 0.446 on the trained checkpoint), and the MWU defender mixes four policies with regret $R(T) \leq \sqrt{T \ln |S|}$, $|S| = 4$. Constants come from the trained reference implementation on a transparent synthetic corpus and are held fixed; Table VI lists each with its provenance tag.

C. A worked end-to-end example

To make (1) concrete we trace one number through it. Take the grid topology, whose routes carry $H = 2$ attacker-controlled hops, and the detector setting $\theta = 0.25$ s. *(1) Threshold to budget.* Since $\theta < s = 5$ s, Lemma 1 gives $\Delta(\theta) \leq H\theta = 0.50$ s of undetected staleness. *(2) Budget to perturbation.* With the measured EKF rate $\gamma = 1.37$ m/s, (8) gives $\|\delta u\| \leq 0.69$ m; the kinematic bound would instead give 7.5 m. *(3) Perturbation to tube.* With $\hat{L}_{\text{loc}} = 1.18$ and $T = 1$, the tube radius is $\rho = 0.69\, e^{1.18} \approx 2.25$ m. *(4) Tube to floor.* For any corridor margin $m \geq 2.25$ m the mission is certified to complete; across the margin family $\{2, 5, 10, 20\}$ m the certified floor is positive for all but the tightest 2 m corridor. Under the kinematic bound, step (3) yields $\rho \approx 24.5$ m, exceeding every margin—which is exactly why the empirical calibration, not the theorem, decides the framing. Fig. 6 shows the measured undetected budget feeding step (2), confirming the saturation predicted by Lemma 1. Step (5), the temporal predicate of (2), is satisfied by a wide margin at this operating point: the induced schedule slip is $\Delta(\theta) \leq 0.50$ s, which is under a 10% tolerance for any mission longer than five seconds.

TABLE VI: Instantiated constants and their provenance. The one pending numeric constant is the PAC–Bayes prior/posterior KL, which we flag rather than substitute.

Component	Constant	Provenance
Lipschitz–Grönwall	$\hat{L}_{\text{loc}}$=1.181 $\Rightarrow$ R=0.153; global $\hat{L}$=1.013 $\Rightarrow$ R=0.182	trained ckpt [fixture]
Rand. smoothing	σ=0.25, R_{ℓ_2}=0.44 (measured 0.446)	trained ckpt [fixture]
PAC–Bayes	McAllester on real R_{emp}; numeric KL pending	methods ch. [pending]
MWU regret	$R(T) \leq \sqrt{T \ln 4}$; T=100 $\Rightarrow$ 11.77	exact
Unit bridge (RF classes)	`caf_shift_v2`: 0.45 m (Grönwall) / 1.15 m (RS)	synthetic IQ [TEXBAT pending]
$\theta \mapsto \delta$ (delay class)	γ=1.20–1.37 m/s measured; kinematic 15 m/s fallback	real 3 flights [hover]

Why these corridor margins. The family $\{2, 5, 10, 20\}$ m is not arbitrary; it brackets the lateral buffers that operational approval frameworks already require, which is what makes the certified window comparable to a regulatory quantity rather than to a modelling choice. Under the JARUS Specific Operations Risk Assessment (SORA), the reference methodology adopted by EASA and other authorities for approving UAS operations in the *specific* category, the operational volume comprises the flight geography plus a *contingency volume* whose minimum lateral extent is the sum of independent error terms: GNSS position accuracy, position-holding error, map error, a reaction-distance term, and the stopping distance of the contingency manoeuvre [45], [46]. SORA v2.5's default values for the first three alone—3 m GNSS accuracy, 3 m position-holding error, 1 m map error—already total 7 m before any reaction or manoeuvre term is added. Our margin family therefore spans the regulator's own scale: 2 m is *tighter* than any SORA-compliant lateral buffer and functions as a deliberately adversarial stress case; 5 m sits inside the default static budget; 10 m is representative of a small multirotor's full contingency extent once reaction distance is included; and 20 m covers a fixed-wing platform, whose larger turn radius and higher cruise speed inflate the manoeuvre term. Reading Fig. 8 against these values converts the abstract question "is θ certifiable?" into the operational one an approval submission must answer: whether the detector setting keeps the worst-case evading tube inside the contingency volume the operator has already declared.

D. The certified floor and the interface that decides it

Fig. 7 is the headline result, and it is deliberately *parametric in the interface mapping*. Under the kinematic worst case, $\theta = 0.25$ s falls just outside the certified window; under the empirically measured mapping it falls *inside* at every corridor

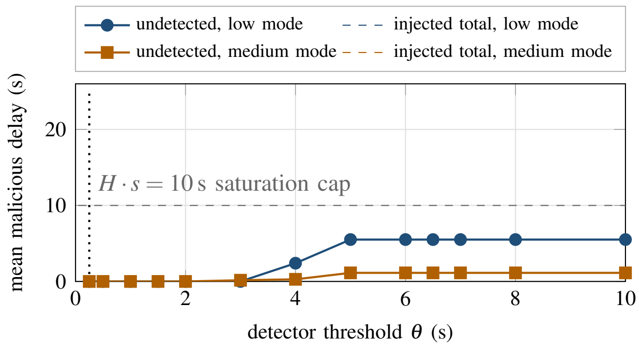

Fig. 6: The measured residual budget behind Lemma 1 (scenario 1, mean over 8 seeds $\times$ 4 policies). The *undetected* delay an evader slips past the detector (solid) saturates well below the *total* injected delay (dashed): beyond the contact-window slack the extra delay produces missed windows flagged at any θ, so the curve the certificate consumes flattens rather than growing.

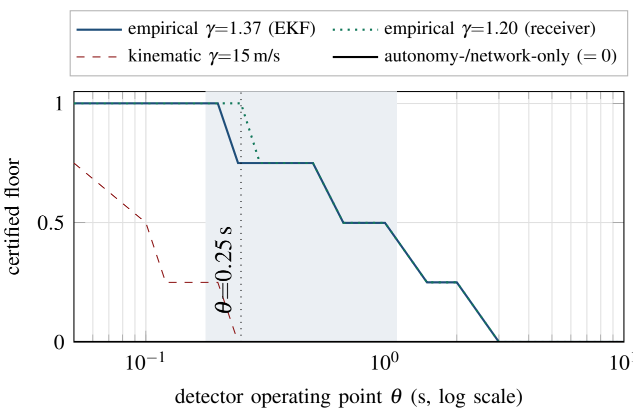

Fig. 7: The headline result: certified floor vs. detector operating point θ for the time-delay class (grid topology, Grönwall tube at the measured local L=1.181, H=2, corridor-margin family $\{2, 5, 10, 20\}$ m). Under the kinematic worst case (red) θ=0.25 s is just outside the certified window; under the empirically measured mapping from real GPS-spoof/jam flights (γ≈1.2–1.4 m/s) it is *inside* at every margin of 5 m or wider (shaded: the EKF window $[0.178, 1.124]$ s at m=10 m); at the tightest 2 m corridor the tube marginally exceeds the margin (2.22 against 2.00 m). Both single-layer baselines certify a zero floor for an evading attacker (black).

margin. Fig. 8 quantifies how much room there is: the mapping would have to steepen past $\gamma_{\text{req}} = 6.14$ m/s (at a 10 m corridor) to push the operating point back out—more than five times the measured rate. This is the paper's most consequential finding for the field: what determines whether cross-layer certification is useful is a *measurement* of how attacks propagate into state error, not the proof machinery.

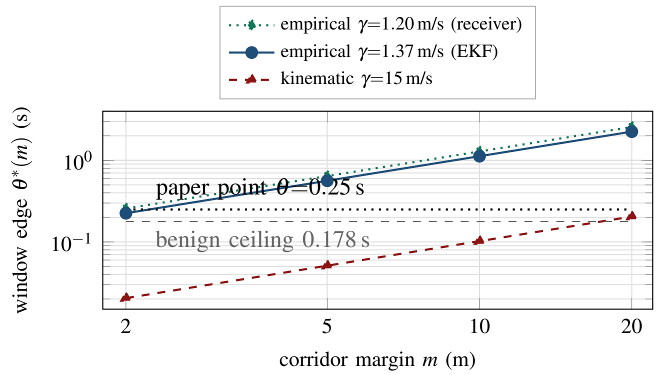

Fig. 8: The decisive sensitivity: the loosest detector setting $\theta^*(m) = m/(\gamma e^{LT} H)$ whose worst-case evading tube still fits a corridor of margin m, per interface mapping. A mapping certifies the paper operating point iff its curve lies above the θ=0.25 s line; the feasible region is bounded below by the measured benign residual ceiling (0.178 s, the FPR=0 floor). The measured rates clear the requirement γ_{req}=6.14 m/s at m=10 m by ∼4.5×; the kinematic worst case fails at every margin.

E. Dominance over single-layer baselines

We state the dominance claim precisely rather than as blanket range-coverage. The composed guarantee dominates two baselines: (i) *autonomy-only*—a navigation certificate with no detector, which must assume the full unmitigated attack including the missed-window penalties (measured cumulative delay up to 424 s) and therefore certifies a zero floor against an evading attacker; and (ii) *network-only*—a detector with no navigation guarantee, for which any evading attack leaves the aircraft undefended, floor zero by definition. **The composed guarantee is the only configuration with a nonzero certified floor at all.** Quantifying per seed with the Wilcoxon signed-rank test and Holm correction, the composed floor exceeds the autonomy-only floor at all 13 operating points on the θ grid, every adjusted $p < 10^{-20}$. The composition wins because it alone uses the detector to *shrink* the perturbation the certificate must absorb, and—via the tightened budget of Lemma 1—the floor saturates rather than collapsing past the knee.

The honest scope: the certified operating window is nonempty but bounded. Under the empirically calibrated interface it is $[0.178, 1.12]$ s at a 10 m corridor, with the operating point $\theta = 0.25$ s comfortably inside; under the conservative kinematic mapping it shrinks to $[0.178, 0.243]$ s and $\theta = 0.25$ s falls just outside. We claim the former and show the latter as the worst-case fallback; we do not claim coverage across the entire relaxation range.

F. Certified-versus-empirical validation, and the two floors

The certificate is sound iff the certified floor never exceeds the realised MCR (Proposition 1). At the anchor $J/S = 20$ dB the strongest defence attains empirical MCR = 0.938 over 93,600 real benchmark flights, comfortably above the certified

TABLE VII: A representative cross-layer pairing: a detector-evading time-delay attack still induces measurable navigation staleness. Values from the released artifact (harness simulation; three-flight δ calibration; benchmark MCR).

Quantity	**Value at representative θ**
Detector threshold θ	0.25 s (paper operating point)
Detector verdict on tuned attacker	evaded (by construction)
Malicious hops H	2 (grid/ring/double-ring routes)
Residual budget $\Delta(\theta) \leq H \min(\theta, s)$	≤ 0.50 s (s=5 s; measured cap 4.84 s)
Measured undetected delay (192 runs)	median 0.00 s, max 0.00 s
Staleness, empirical γ=1.37 m/s	≤ 0.69 m (Whelan, hover)
Staleness, kinematic $v_{\max}$=15 m/s	≤ 7.5 m (worst case)
Paired autonomy window	UAV-EW-BENCH at J/S=20 dB
Empirical MCR of paired window	0.938 (strongest defence)
`pairing_basis`	synthetic / measured (per record)

target 0.80; the gap is the conservativeness of the worst-case Grönwall bound, and it narrows where the empirically tightened interface replaces the kinematic worst case.

We keep two floors distinct throughout, because conflating them would overstate the guarantee. The reference certificate targets a *certified* MCR ≥ 0.80 at the anchor operating point; the empirical benchmark reports the DO-326A 0.90 *operational* floor holding further into the band. These are different quantities—a conservative proof-derived bound versus a measured rate—and we label every reported number accordingly.

G. Three evidence classes, one worked pairing each

Table VII reports the pairing at a representative operating point. The `pairing_basis` flag is easiest to understand through one example of each class. *Measured, same platform:* the UAV ATTACK DATASET records a real GPS-jamming flight and its measured position error on one airframe; a pairing joining that flight to the corresponding RF event is tagged `measured_same_platform`, and its δ is a measurement, not a model. *Physics model:* a forwarding-attack window carries a residual budget in seconds; coupling it to a navigation window through the stated relation (8) yields a `physics_model` pairing—a named, versioned physical relation a reviewer can inspect and challenge, and the class the certificate consumes. *Synthetic alignment:* when a network window from one simulated source and an autonomy window from another are aligned by construction, the pairing is tagged `synthetic_alignment`: maximal coverage, weakest causal claim, and the flag says so. A user needing a strong claim filters to the first class; a user needing breadth accepts the third with its provenance explicit.

H. How much of the benchmark rests on measurement

The flag is only useful if its distribution is published, so we report it rather than describe it. Table VIII gives the acquisition split across the whole corpus: 200,610 of 431,773 ingested events (46.5%) were captured on real hardware, the remainder generated by simulators of three different kinds. That split is favourable, but it is also the *weaker* of the two questions a reviewer should ask, and conflating them would overstate the benchmark.

The stronger question is how many *pairings* rest on measurement, and here the honest answer is more restrictive. A `measured_same_platform` pairing requires one source to have observed both the network event and the flight it degraded on a single airframe. Exactly one source in the corpus can do this—the UAV ATTACK DATASET—and although its three flights supply the delay-to-position rate γ that the certificate consumes (Table IV, a real-corpus measurement obtained by self-referencing each flight to its own pre-attack median), the release does not carry machine-readable attack on/off timestamps. Without them the adapter cannot emit the `AttackWindow` boundaries a pairing record requires, and we decline to infer them. **The number of released `measured_same_platform` pairings is therefore currently zero**: every released pairing is `physics_model` (the network window coupled to a navigation window through the stated relation (8), and the class the theorem consumes) or `synthetic_alignment` (aligned by construction under Appendix C, with the error budget (16)).

We state this plainly because it is exactly the fact the flag exists to expose, and because it locates the benchmark's binding limitation precisely: not the volume of data, of which there is a great deal, but the scarcity of captures that observe both layers at once. The headline guarantee does not depend on closing that gap—γ is measured on real flights regardless of whether the window boundaries are machine-readable—but the strongest evidence class the schema can express is, today, unpopulated, and a reader filtering the artifact to `measured_same_platform` will correctly retrieve nothing. Obtaining same-platform captures with published attack intervals, across flight regimes rather than hover alone, is consequently the highest-value extension of the corpus, and §XII treats it as such.

XI. DISCUSSION

What the composition buys an operator. It turns a network-security knob into a navigation-safety number. A fleet operator setting a detector's false-positive budget currently reasons in packet statistics—"how much benign traffic can I afford to flag?"—with no way to connect that choice to airworthiness. Theorem 1 supplies the missing translation: each detector setting maps to a certified mission floor, so the operator can instead ask "what detector tightness do I need to certify my corridor?" and read the answer off Fig. 8.

Why the interface, not the theorem, is decisive. A reader might expect the certificate to be the hard part. Our results say otherwise: the tube is a short consequence of a

TABLE VIII: Acquisition provenance across the ingested corpus. Counts are from the schema-validated adapter outputs; the split is machine-generated by `experiments/provenance_ledger.py` so the paper cannot drift from the artifact. Acquisition is a property of the *source*; the strictly stronger `pairing_basis` distribution is discussed in the text.

Source	**Acquisition**	**Events**	**%**
HCRL_UAVCAN	real testbed	200,000	46.3
UAVIDS_2025	simulation (NS-3)	122,171	28.3
UAV_EW_BENCH_2026	simulation (physics-informed)	93,600	21.7
DATAMUT_SIM	simulation (event)	15,392	3.6
UAV_ATTACK_WHELAN	real testbed	610	0.1
UAV_CAS	simulation (digital twin)	—	—
real testbed	captured on hardware	**200,610**	**46.5**
simulation	generated	**231,163**	**53.5**

classical inequality and the Lipschitz constant is measured, not assumed. What determines whether a useful detector setting is certifiable is γ, the metres-per-second of position error a second of staleness buys the attacker. The order-of-magnitude gap between the kinematic bound and the measured rate is the difference between a vacuous certified window and one that contains the operating point. This relocates the research agenda: tightening cross-layer guarantees is primarily a *measurement* problem about how attacks propagate into state error—which is precisely why the benchmark and the theorem belong in one paper.

Two methodological findings we surface rather than bury. Building the benchmark forced two corrections that are instructive for anyone assembling a cross-layer corpus. First, our initial operating curve was contaminated by an accumulation bug in the sweep harness, in which each operating point inadvertently averaged all previous points; the tell was recall *rising* with the threshold in one regime, which is impossible for a threshold detector. The clean multi-seed campaign in Fig. 3 replaces it. Second, the bus dataset's attack labels are not machine-readable per scenario, so rather than guess we derived them from frame statistics and flagged the derivation for confirmation. Both episodes argue *for* the provenance discipline the schema enforces: a benchmark that records how every number was obtained lets errors be caught rather than propagated.

Generality. Although we instantiate on a time-delay forwarding attack and a GNSS navigation model, the chain is agnostic to both ends. Any detector with a monotone operating point and any perturbation-to-state mapping slot into Lemma 1 and (8); the theorem and the four-certificate cover are unchanged. Extending the interface catalogue to jamming-intensity-to-error and to bus-level attacks is direct future work.

XII. Limitations and Future Work

We state limitations plainly; none undercuts the contributions, and each defines a concrete extension.

The delay-to-position rate is a three-flight hover calibration. The decisive quantity γ is measured on three real flights, all in a low-speed hover/loiter regime; at cruise speed the position error per second of staleness could be larger. This is the single result that could change the paper's framing: if a cruise-inclusive calibration measured $\gamma > \gamma_{\text{req}} = 7.3$ m/s at a 10 m corridor—more than a five-fold increase over what we observe—the operating point would leave the certified window and the honest framing would revert to the narrow "certify-small, detect-large" regime that the conservative kinematic bound already guarantees. A larger, cruise-inclusive flight campaign is the highest-value follow-up.

Conservatism of the deterministic core. The Grönwall bound grows as e^{LT}, so for long horizons or higher Lipschitz constants the certified floor falls below what is empirically observed. This is by design—it is why the architecture is a four-certificate cover—but fusing the deterministic, smoothing, and PAC-Bayes floors into a single tighter envelope is open.

Evidence-class imbalance. Of the six datasets only one couples a network/RF event to a measured flight on one airframe, so it is the sole source of `measured_same_platform` pairings; the others support physics-modelled or synthetically-aligned couplings. This honestly reflects what public data exists, and is exactly why the honesty flag is mandatory rather than optional.

Calibration constants that remain data-gated. The unit bridge mapping a raw GNSS position error into receiver-level feature space (for the RF-manipulation classes, as opposed to the time-delay class the theorem targets) is calibrated on a synthetic signal corpus pending access to real in-phase and quadrature recordings; and the PAC-Bayes bound (14) is wired to the trained model's measured empirical risk but awaits its numeric KL divergence. Neither gates the headline result; both are tracked explicitly so that no synthetic constant is reported as real.

Analytical navigation anchor and detector scope. The benchmark's completion outcomes come from a physics-informed analytical Monte-Carlo backend calibrated to full-fidelity anchors; results should be described as a calibrated analytical benchmark, and we do so. We also instantiate the harness on one forwarding detector whose operating point is a single clean threshold, chosen because it makes the cross-layer sweep unambiguous; extending to multi-dimensional operating points requires a higher-dimensional sweep, not a schema change.

Threats to validity. Construct: does a recorded pairing represent the causal link it claims? The honesty flag bounds this risk, and a reviewer can restrict any analysis to the measured class. *Internal:* are the numbers correct? Two of our own measurement errors were caught precisely because the provenance discipline made them visible, and every figure is regenerated from committed data. *External:* four of six sources are simulation or calibrated twin, so absolute detector-performance numbers are corpus-specific; what transfers is the *structure*—the knee-and-floor operating curve and the order-of-magnitude gap between kinematic and empirical staleness—which is a property of store-and-forward scheduling and estimator behaviour, not of any one simulator.

XIII. Conclusion

UAV security has been studied as two separate problems—attacks on the wire and attacks on the flight—because nothing let it be studied as one. We built the benchmark that makes the coupling representable and then proved, on exactly its objects, the guarantee that the coupling enables. The benchmark contributes a two-layer schema whose unit is the cross-layer pairing, six validated adapters, and a detector-operating-curve harness; the guarantee contributes a residual budget that measurement tightens into a *saturating* bound $\Delta(\theta) \leq H \min(\theta, s)$, an interface rate calibrated on real spoofing and jamming flights, and a composition theorem chaining a detector's operating point to a certified mission-completion floor.

Three findings survive scrutiny. First, the composed guarantee is the only configuration that certifies a nonzero mission floor against a detector-evading attacker, dominating both single-layer baselines at every operating point with overwhelming statistical significance. Second, the empirically calibrated interface—an order of magnitude tighter than the kinematic worst case—brings the detector's real operating point inside the certified window, so the guarantee is operationally relevant and not merely formal. Third, and most consequential for the field, the binding constraint on cross-layer certification is the measurement of how attacks propagate into state error, not the proof machinery: the interface, not the theorem, decides the outcome. UAV network security and UAV autonomy security, we conclude, are not two problems but two ends of one certificate.

XIV. Ethics Considerations

This work involves no human subjects and no personally identifying data. All six datasets are public research corpora released by their authors for security research; we redistribute none of the raw third-party data, shipping only integration scripts and unified labels so that each source is fetched from its origin under its own licence. No live aircraft, production network, or third-party system was attacked: every attack in this paper is either replayed from a published capture or generated in simulation on infrastructure we control.

The dual-use question deserves a direct answer. Our operating-curve harness quantifies how an attacker can stay under a detector's threshold, which is information an adversary could in principle use. We judge the disclosure clearly net-positive and consistent with standard practice in intrusion-detection research, for three reasons. First, the evasion strategy is not novel—staying under a published threshold is the definitional behaviour of an evading attacker, already assumed in the detector's own analysis. Second, our central result *bounds* what such an attacker achieves and shows it saturates, which favours defenders. Third, the practical consequence we identify is a defensive configuration procedure: read the required detector tightness off Fig. 8 for your corridor margin. We disclose no previously unknown vulnerability in a deployed system, so no coordinated-disclosure process was applicable; the underlying detector's authors are aware of and have commented on this analysis of their operating point.

XV. Open Science and Artifact Availability

Everything needed to reproduce this paper is released as an open artifact: the schema, the six ingestion adapters, the operating-curve harness, the certificate engine, the statistical analysis, and the figure generator. Every figure and table in this paper is machine-generated from committed result files rather than transcribed, so a reader can regenerate them from the released data; each carries a provenance tag (real-corpus, simulation-derived, or stated modelling parameter) in an artifact ledger. The harness is deterministic per seed, and headline comparisons are released per seed with their nonparametric tests. We intend to submit the artifact for evaluation. The repository is anonymised for review and will be de-anonymised on acceptance.

Appendix

The composition proved in this paper spans two research communities that share very few conventions: one reasons about packets and detector thresholds, the other about trajectories and certified radii. This section fixes the single system model both halves of the argument refer to—how an autonomous UAV closes its navigation loop, which of its interfaces an adversary can reach, and what a robustness *certificate* does and does not promise—so that the interface mapping of §VII joins two precisely defined quantities rather than two informal ones. Acronyms are defined at first use and not tabulated; Table IX is the single reference for mathematical symbols. A reader fluent in both UAV autonomy and certified robustness may proceed directly to §II.

A. How an autonomous UAV stays on course

An unmanned aerial vehicle (UAV, colloquially a drone) closes a control loop around a *state estimate*. An onboard computer fuses fixes from a Global Navigation Satellite System (GNSS, of which the American GPS is one constellation) receiver, inertial measurements from accelerometers and gyroscopes (the IMU), and barometric height into a best guess of where the aircraft is and how fast it is moving. On the PX4 autopilot family used throughout this paper the fusion is performed by an extended Kalman filter (EKF)—a standard recursive estimator that weighs each sensor by its expected noise [39]. The flight controller steers to keep the estimate on the planned route.

Two properties of this loop drive everything that follows. First, the loop is only as good as its *freshest trustworthy input*: if GNSS is denied, the EKF "coasts" on inertial data and its position error grows with time. Second, the loop is *networked*. Swarm members and the ground control station exchange corrections and commands over radio links: a command-and-control (C2) link, typically speaking the Micro Air Vehicle Link (MAVLink) protocol, a multi-hop mesh in which peers relay each other's traffic during periodic contact windows—a delay-tolerant network (DTN)—and an intra-vehicle bus—the UAV Controller Area Network (UAVCAN, also DroneCAN)— connecting the flight controller to motor controllers and sensors. An attacker on the mesh can therefore reach the navigation loop without ever touching the radio-frequency (RF) front end.

B. The two attacker technologies

Electronic warfare (EW) attacks the sensing physics. A *jammer* raises the jamming-to-signal power ratio (J/S, in decibels) until the receiver loses satellite lock; a *spoofer* broadcasts counterfeit satellite signals so the receiver locks onto a false position [47], [48]. *Network attacks* target the messages: a *time-delay attack*—the class our theorem covers—has a compromised relay hold packets before forwarding them, so downstream consumers act on stale state [2], [3]. Neither needs to break cryptography; both degrade the control loop through legitimate-looking inputs, which is precisely why detection and robustness must be composed rather than assumed.

C. What a "certificate" promises

A *certified* robustness statement differs from an empirical evaluation: instead of reporting that attacks in a test set were survived, it proves that *every* perturbation within an explicitly stated budget is survived. The price is conservatism—certified bounds are worst-case, so a certified floor typically sits below what is observed empirically. We keep the two kinds of number separate and label them throughout: *certified* floors come from proofs, *empirical* rates from measured flights. We use two flavours of certificate: deterministic Lipschitz-type bounds (if a model cannot amplify input changes by more than a factor L, a bounded input change yields a bounded output change) and probabilistic randomized-smoothing bounds (adding calibrated noise and taking a majority vote yields a radius within which the decision provably cannot flip) [17]. The regulatory anchor for "mission survived" is the airworthiness-security process standard DO-326A/ED-202A [4], whose industry-acceptable completion floor of 0.90 appears in our figures as a horizontal reference line.

Because the sources share no global clock, a `synthetic_alignment` pairing must state exactly how a network window and an autonomy window from different captures are placed on a common axis, and how much error that placement can introduce. Left implicit, this is the step at which a cross-layer benchmark could silently manufacture—or destroy—the very effect it reports.

Let the network source have relative clock $\tau_n \in [0, T_n]$ and the autonomy source $\tau_a \in [0, T_a]$, each measured from its own capture start. An alignment is an affine map

$$\phi(\tau_n) \;=\; a\,\tau_n + b, \qquad a = \frac{T_a}{T_n}, \tag{15}$$

whose scale a normalises the two capture durations and whose offset b is fixed by *anchoring attack onsets*: we set b so that $\phi(t_0^n) = t_0^a$, where t_0^n is the first labelled malicious event in W_n and t_0^a the onset of degradation in W_a. Onset is not eyeballed: t_0^a is the first sample at which the autonomy metric exceeds its pre-attack median by more than $3\hat{\sigma}$, with $\hat{\sigma}$ the robust (median-absolute-deviation) scale of the pre-attack segment. Both anchors are therefore defined by the source labels and the source's own noise floor, not by the analyst.

Three error terms bound the residual misalignment e_{align}:

$$e_{\text{align}} \;\le\; \underbrace{\tfrac{1}{2}(\Delta t_n + \Delta t_a)}_{\text{quantisation}} \;+\; \underbrace{|a-1|\,T_a}_{\text{rate mismatch}} \;+\; \underbrace{e_{\text{onset}}}_{\text{anchor detection}}, \tag{16}$$

where $\Delta t_n, \Delta t_a$ are the two sources' sampling intervals (a per-flow network record and a 50 Hz telemetry stream give $\frac{1}{2}(\Delta t_n + \Delta t_a) \le \frac{1}{2}(1.0 + 0.02) = 0.51$ s), the rate term vanishes when both captures are replayed at their native rate ($a = 1$, the case for every pairing we release), and $e_{\text{onset}} \le \Delta t_a$ is the one-sample latency of the $3\hat{\sigma}$ crossing. For the released `synthetic_alignment` pairings this gives $e_{\text{align}} \le 0.53$ s.

This bound is what keeps the alignment from biasing the reported staleness. The quantity the certificate consumes, $\Delta(\theta)$, is computed *entirely within the network source* from per-hop residual delays; the alignment determines only which autonomy window a given network window is paired *with*, never the numeric value of $\Delta(\theta)$. Misalignment can therefore attach a pairing to the wrong autonomy interval, but it cannot inflate or deflate $\Delta(\theta)$ itself. Its effect is confined to the empirical MCR column, and it is bounded there too: since

TABLE IX: Mathematical notation. Every symbol is defined where first used; this table is the single reference.

W_n, W_a	network / autonomy attack window
P	cross-layer pairing tuple, (4)
$x(t) \in \mathbb{R}^n$	navigation state (position, velocity, …) at time t
$u(t)$, δu	sensor input to the navigation dynamics; its perturbation
F	drift (right-hand side) of the navigation dynamics (10)
T	mission horizon (normalised to 1 in the instantiation)
m	safe-corridor margin: allowed lateral deviation, metres
D_θ, θ (ε)	network detector; its operating point (tolerated per-hop residual delay, s)
s, P_c	contact-window slack (5 s); contact period (60 s)
H	number of attacker-controlled (malicious) hops on a route
$\Delta(\theta)$	residual budget: worst undetected cumulative delay, s
$\delta(\theta)$	induced perturbation on the navigation input
γ	delay-to-position rate: metres of error per second of degradation
$v_{\max}$	kinematic worst-case speed bound (15 m/s)
L, $\hat{L}$, $\hat{L}_{\text{loc}}$	(estimated / local) Lipschitz constant of F
$\rho(\theta)$	trajectory-tube radius $\delta(\theta)e^{LT}$, metres
$\theta^*(m)$	loosest θ whose tube fits margin m (window edge)
$f(\cdot)$	certified MCR floor as a function of the perturbation budget
R, σ, α, n	certified input radius; smoothing noise, confidence, samples
R_{emp}, KL	empirical risk and prior-posterior divergence, (14)
$\lvert S\rvert$	size of the defender's policy set in the MWU certificate (4)

$e_{\text{align}} \leq 0.53\,\text{s}$ is two orders of magnitude below the $60\,\text{s}$ contact period that separates distinct attack windows, no released pairing can be displaced onto a neighbouring window. Pairings whose alignment error would exceed one tenth of the shorter window's duration are rejected by the adapter rather than emitted with a caveat.

Proof of Theorem 1. Fix a mission and let $x_{\text{nom}}(t)$ be the nominal trajectory—the solution of (10) under the attack-free input $u(t)$—and $x_{\text{att}}(t)$ the trajectory under $u(t)+\delta u(t)$, both started from $x_{\text{nom}}(0) = x_{\text{att}}(0)$. Write $e(t) = x_{\text{att}}(t) - x_{\text{nom}}(t)$. From (10), $\dot{e}(t) = F(x_{\text{att}}, u+\delta u) - F(x_{\text{nom}}, u)$, so L-Lipschitzness gives $\|\dot{e}(t)\| \leq L\|e(t)\| + L\|\delta u(t)\|$. Integrating with $e(0) = 0$,

$$\|e(t)\| \;\leq\; L\int_0^t \|e(\tau)\|\,d\tau \;+\; L\int_0^t \|\delta u(\tau)\|\,d\tau.$$

Applying the integral form of Grönwall's inequality [40] to the non-decreasing forcing term $g(t) = L\int_0^t \|\delta u\|\,d\tau$ yields $\|e(t)\| \leq g(t)e^{Lt}$, and bounding the input by $\|\delta u(\tau)\| \leq \delta(\theta)$ gives, at the horizon,

$$\|e(T)\| \;\leq\; \delta(\theta)\big(e^{LT}-1\big) \;\leq\; \delta(\theta)e^{LT}, \tag{17}$$

which is (11); we report the closed-form right-hand bound. Now suppose $\rho(\theta) \leq m$. The nominal trajectory lies in the safe corridor by construction, and every point of the attacked trajectory is within $\rho(\theta) \leq m$ of it by (17); hence the attacked trajectory never exits the corridor over $[0,T]$, discharging the spatial predicate of (2). For the temporal predicate, the only mechanism by which a D_θ-evading time-delay attack displaces arrival is the undetected forwarding delay it accumulates, which Lemma 1 bounds by $\Delta(\theta)$; hence $t_{\text{arr}} \leq T + \Delta(\theta)$, and $\Delta(\theta) \leq \kappa T$ gives $t_{\text{arr}} \leq (1+\kappa)T$ as required. Both predicates holding, the mission completes. Taking the probability of $\{\rho(\theta) \leq m\} \cap \{\Delta(\theta) \leq \kappa T\}$ over the mission distribution—the only source of randomness once θ is fixed, since $\delta(\theta)$ and $\Delta(\theta)$ are deterministic bounds—gives (12). Composing $\theta \mapsto \Delta(\theta)$ (Lemma 1), $\Delta \mapsto \delta$ (§VII), and $\delta \mapsto \text{MCR}$ establishes the chain (1). □

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